1. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

Relevance: ★★★★★ (Highest)

- Students must correctly interpret multiple assumptions:
  - Equal numbers of males and females forming couples at the beginning of each year
  - Each couple raises one chick per year
  - A 20% mortality rate applied at the end of each year
  - One-year-old penguins also reproduce
- These statements must be **integrated into a coherent dynamic model**

👉 Any misunderstanding leads to an incorrect choice of formula.

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2. Identify the constraints, assumptions and simplifications in a mathematical model

Relevance: ★★★★★

- The task is essentially about **selecting a model under fixed assumptions**
- Key modelling assumptions include:
  - A constant 1:1 sex ratio
  - Equal reproductive ability for all penguins
  - An age-independent mortality rate
- The correct formula must **fully respect these idealised conditions**

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3. Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

Relevance: ★★★★☆

- Each year follows the same structure:
  - Population × 1.5 (reproduction)
  - Then × 0.8 (survival after mortality)
- This represents a **constant proportional change per year**
- Mathematically, this is an **exponential (recursive) relationship**

4. Recognise aspects of a problem that correspond with known mathematical concepts, facts or procedures

Relevance: ★★★★☆

- The situation corresponds to:
  - Exponential growth models
  - Geometric sequences
  - Classic "growth × survival rate" population models
- Students must connect the context to the general form
  $P_n = P_0 \cdot r^n$

5. Translate a problem into a standard mathematical representation or algorithm

Relevance: ★★★★☆

- From the description, one can form the recurrence:

$$P_{n+1} = P_n \times 1.5 \times 0.8$$

- Which generalises to:

$$P = 10\,000 \times (1.5 \times 0.8)^7$$

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6. Select a mathematical description or a representation that describes a problem

Relevance: ★★★☆☆

- Students are not required to derive a formula from scratch, but must:
  - Compare different exponential expressions
  - Select the one that best matches the situation

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7. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

Relevance: ★★★☆☆

- Key mathematical elements include:
  - Total population P
  - Time in years as an exponent
  - Growth and survival factors as parameters

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8. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

Relevance: ★★★☆☆

- The yearly process must be decomposed into:
  1. A reproduction phase
  2. A mortality phase
- Because the structure repeats annually, it can be represented using powers

9. Identify the key variables in a model

Relevance: ★★★☆☆

- Important variables include:
  - Initial population (10,000)
  - Number of years (7)
  - Annual growth factor (1.5)
  - Annual survival factor (0.8)

① Read, decode and make sense of statements… to create a model of the situation（理解题意并建立模型）

相关性：★★★★★（最高）

- 学生必须正确理解多条文字假设：
  - 每年年初成对
  - 每对每年 1 只幼崽
  - 每年年底死亡率 20%
  - 一岁企鹅也参与繁殖
- 需要将**分散的自然语言信息整合成一个连贯的动态模型**

👉 若任一条件理解错误，公式判断必然出错。

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② Identify the constraints, assumptions, simplifications in a mathematical model（识别模型的假设与约束）

相关性：★★★★★

- 本题本质上是**"在固定假设下选模型"**
- 明确的建模假设包括：
  - 性别始终均等
  - 所有个体生育能力相同
  - 死亡率不随年龄变化
- 正确公式必须**完全符合这些理想化条件**

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③ Recognise mathematical structure (regularities, relationships, patterns)（识别数学结构）

相关性：★★★★☆

- 每一年都重复同一结构：
  - 数量 × 1.5（繁殖）
  - 再 × 0.8（死亡）
- 体现出**固定比例的年变化率**
- 本质是一个**指数型递推关系**

④ Recognise aspects of a problem that correspond with known mathematical concepts or procedures（识别熟悉的数学模型）

相关性：★★★★☆

- 对应：
  - 指数增长模型
  - 几何数列
  - "增长 × 存活率"的经典人口模型
- 能将情境与"$P_n = P_0 \cdot r^n$"结构建立联系

⑤ Translate a problem into a standard mathematical representation or algorithm（转化为标准数学表达）

相关性：★★★★☆

- 从语言描述得到：

$$P_{n+1} = P_n \times 1.5 \times 0.8$$

- 再推广为：

$$P = 10\,000 \times (1.5 \times 0.8)^7$$

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⑥ Select a mathematical description or a representation that describes a problem（选择合适的数学表述）

相关性：★★★☆☆

- 学生不需要自行写公式，但需在选项中：
  - 比较不同指数模型
  - 判断哪一个最符合情境

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⑦ Identify and describe the mathematical aspects of a real-world problem situation（识别现实问题中的数学要素）

相关性：★★★☆☆

- 明确：
  - 总企鹅数是变量 P
  - 年数是指数
  - 增长率和存活率是核心参数

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⑧ Simplify or decompose a situation to make it amenable to mathematical analysis（分解情境）

相关性：★★★☆☆

- 将一年内变化拆分为：
  1. 繁殖阶段
  2. 死亡阶段
- 每年结构相同 → 可用幂表示

⑨ Identify the key variables in a model（识别关键变量）

相关性：★★★☆☆

- 关键变量包括：
  - 初始种群数量（10,000）
  - 年数（7）
  - 年增长因子（1.5）
  - 存活因子（0.8）